Adaptive Morphology Filter: A Lightweight Module for Deep Hyperspectral Image Classification

Fei Zhou, Xin Sun, *Member, IEEE*, Chengze Sun, Junyu Dong, *Member, IEEE*, and Xiao Xiang Zhu, *Fellow, IEEE*

***Abstract*— Deep neural network models significantly outperform classical algorithms in the hyperspectral image (HSI) classification task. These deep models improve generalization but incur significant computational demands. This article endeavors to alleviate the computational distress in a depthwise manner through the use of morphological operations. We propose the adaptive morphology filter (AMF) to effectively extract spatial features like the conventional depthwise convolution layer. Furthermore, we reparameterize AMF into its equivalent form, i.e., a traditional binary morphology filter, which drastically reduces the number of parameters in the inference phase. Finally, we stack multiple AMFs to achieve a large receptive field and construct a lightweight AMNet for classifying HSIs. It is noteworthy that we prove the deep stack of depthwise AMFs to be equivalent to structural element decomposition. We test our model on five benchmark datasets. Experiments show that our approach outperforms state-of-the-art methods with fewer parameters ($\approx 10k$). The codes will be publicly available at https://github.com/zhu-xlab/Adaptive-Morphology-Filter.**

***Index Terms*— Deep learning (DL), hyperspectral image (HSI) classification, morphology filter, structural reparameterization (SRP).**

I. Introduction

A HYPERSPECTRAL image (HSI) captures hundreds of contiguous and narrow spectral bands, recording landform information by aerial sensors [1]. Over the past decades, HSI has been of great interest to a wide variety of fields, e.g., urban surface classification, precise agriculture, biomedical exploration, and resource exploration [2], [3]. HSI classification requires fast and real-time responses, especially on edge devices such as drones and satellites [4], and therefore rapid inference speed is necessary. In recent years, deep learning (DL) methods [5], [6], [7] have achieved rapid development in the HSI analysis task and are superior to classical machine learning. However, the overparameterization [8], [9], [10], [11] nature makes DL-based methods challenging to employ on the edge devices.

To tackle the problem, many approaches are proposed. Some recent methods focus on the compact network design (CND). Those methods reduce network parameters by employing some compact modules, depthwise convolution [12] reduces computational overhead by eliminating channel interactions within convolutional layers, while MDCN [13] enhances performance by incorporating multiscale depthwise convolution. LS^{2}CM [14] proposes a combination of depthwise convolution and spectral–spatial features for hyperspectral classification. The ghost block technique minimizes computation by leveraging redundant features between channels [15]. Paoletti et al. [16] were the first to introduce the ghost block in hyperspectral classification tasks, while lightweight spectral–spatial attention feature fusion network (LMAFN) [17] integrates the ghost block with an attention mechanism to improve performance. The shuffle operation, on the other hand, augments the efficacy of depthwise convolution by enhancing dimension interactions. Wang et al. [18] incorporated the shuffle operation to enhance attention models, thereby strengthening spectral–spatial feature representation. Lightweight feature pyramid module (LWFPM) [19], meanwhile, employs the shuffle operation to blend features of different scales, leading to enhanced performance.

The model structure and processing pipeline of CND are consistent during training and testing. Despite their success, those methods are still resource-intensive and complex for edge devices in general. The parameter reduction of these compact models can cause instability during the training process. The other two candidates are weight quantization (WQ) [20], [21] and model pruning (MP) [22], [23], [24]. WQ trains the network in floating-point and replaces the floating-point operation with an integer operation in the test phase, while MP removes redundant structures and parameters based on the sparsity of the network after training. However, both of these two solutions will result in performance deterioration in the inference phase. Structural reparameterization (SRP) possesses the advantages of multibranch structure during training, ensuring network performance. Simultaneously, it accelerates inference during testing without sacrificing accuracy. This

Manuscript received 18 July 2023; revised 1 October 2023; accepted 18 October 2023. Date of publication 25 October 2023; date of current version 7 November 2023. This work was supported in part by the National Natural Science Foundation of China under Project 61971388 and in part by the Alexander von Humboldt Foundation. The work of Xiao Xiang Zhu was supported in part by the German Federal Ministry of Education and Research (BMBF) through the Framework of the International Future AI Lab "AI4EO–Artificial Intelligence for Earth Observation: Reasoning, Uncertainties, Ethics and Beyond" under Grant 01DD20001, in part by the German Federal Ministry for Economic Affairs and Climate Action through the Framework of the "National Center of Excellence ML4Earth" under Grant 50EE2201C, and in part by the Munich Center for Machine Learning. *(Corresponding author: Xin Sun.)*

Fei Zhou, Chengze Sun, and Junyu Dong are with the College of Information Science and Engineering, Ocean University of China, Qingdao, Shandong 266100, China.

Xin Sun is with the Data Science in Earth Observation (SiPEO, former: Signal Processing in Earth Observation), Technical University of Munich (TUM), 80333 Munich, Germany (e-mail: sunxin1984@ieee.org).

Xiao Xiang Zhu is with the Data Science in Earth Observation (SiPEO, former: Signal Processing in Earth Observation), Technical University of Munich (TUM), 80333 Munich, Germany, and also with the Munich Center for Machine Learning, 80333 Munich, Germany.

Digital Object Identifier 10.1109/TGRS.2023.3327418

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

provides a practical way to meet the fast and real-time requirements of HSI classification tasks.

It is critical to deploy HSI classification models on resource-constrained edge devices but difficult due to the large size of deep models. Existing lightweight models either lack attractive performance or suffer from performance losses during inference. Additionally, the limited training sample further hinders the generalization of the deep model, which also poses great challenge to HSI. Previous works [25], [26] demonstrate that the prior knowledge helps network learn discriminative features from few training data. Over the past decade, the traditional morphological filter has achieved great success in HSI classification, e.g., extended morphological profiles (EMPs). We argue that morphological operators, as manually designed feature extractors, can be regarded as a type of structure with innate prior. And its depth-wise form shares similarities with depthwise convolution. However, the disadvantage of non-differentiable binary weights has blocked its exploration in deep neural networks.

To address these challenges, we propose the first differentiable binary morphological filters in HSI classification, adaptive morphology filter (AMF), enabling end-to-end training of binary morphological operator. In addition, we first introduce the SRP technique to transfer AMF to a traditional morphological filter, which effectively reduces network size without compromising test accuracy. Compared to DL-based methods employing similar morphological operations, our approach directly optimizes binary structural elements rather than utilizing fixed structural elements [27], [28], [29] or softened structural elements [30], [31]. The former struggles to capture intricate spatial features, while the latter either incurs a performance loss when transitioning to binary weights or fails to effectively reduce computational overhead. In addition, the binary structural element allows us to adopt a decomposition strategy, breaking down the AMF into stacked smaller AMFs, utilizing them as the backbone of our network and delivering multiscale morphological features. This strategy further reduces the network's complexity.

The main contributions are threefold.

1) *AMF:* We propose a learnable AMF to learn the spatial feature in an efficient way (depthwise) and adopt SRP to further shrink the model size. Unlike others, our reparameterization target is a morphology filter rather than conventional convolution.
2) *Adaptive Structural Element Decomposition (ASED):* We revisit the deep architecture of CNN and find that the stack of small depthwise AMFs can be regarded as the decomposition of a large structural element in our paradigm. The decomposition is data-driven and flexible because AMFs can be trained end-to-end.
3) *Adaptive Morphology Network (AMNet):* We propose a lightweight AMNet for classifying HSIs based on AMFs and ASED. It can be deemed a DL version of the EMP technique, and AMNet is patch-free in the inference stage; it avoids redundant pixels in overlap patches and accelerates the inference speed. AMNet delivers the best trade-off between accuracy and model size compared to recent efforts.

The rest of this article is organized as follows. Section II briefly introduces the development of HSI classification and related technologies, e.g., SRP and EMP. In Section III, our AMF, ASED, and AMNet are proposed and discussed. Section IV shows the experimental results and compares them with other related methods. Section V concludes the article.

II. Related Work

In this section, we first give a short description of recent developments in related fields and then briefly review some related concepts, like SRP and EMPs.

A. Recent Development

Due to the presence of hundreds of spectral channels in hyperspectral imaging, the curse of dimensionality becomes a significant concern [32], [33]. To address this problem, early works typically employed dimensionality reduction or channel selection techniques [34], [35], [36]. While these methods consider the interrelationships among spectral channels, they often overlook the extraction of spatial information. Spectral heterogeneity represents another pressing challenge [37], and the spatial information is helpful to solve the problem. As a result, various works [38], [39], [40] have started fusing spatial and spectral information. Among these techniques, morphological filters [41], [42] have achieved outstanding performance. However, the structural elements in morphological filtering are fixed, which makes it challenging for morphological filters to extract spatial information pertaining to arbitrary object shapes in complex scenarios. In our study, we introduce AMF, which, through end-to-end learning of binary structural elements, can extract morphological profiles features of arbitrary shapes, thereby enhancing network performance.

DL methods are widely considered for extensive vision tasks, including HSI classification. For example, contextual deep CNN (CDN) [43] uses multiscale 3-D convolution to extract rich contextual features. Recurrent neural network (RNN) is utilized [44] to process the intrinsic sequence data structure in HSI. Bayesian CNN [45] uses CNN to learn the posterior class distributions and unifies the spectral and spatial information in the Bayesian framework. Some DL methods are inspired by traditional techniques. For instance, DHTCSCNet [46] incorporates the idea of sparse coding, training deep high-order filters to capture spatial–spectral structure information. SSRPNet [47] combines the local binary pattern (LBP) features and uses random patches as the convolution kernels. CNN-enhanced graph convolutional network (CEGCN) [48] combines superpixel techniques and GCN by integrating the graph encoding process into the network, enabling the learning of features at various scales. Deep low-rank graph convolutional subspace clustering (DLR-GCSC) [49] introduces clustering concepts in HSI classification, extracting more discriminative features by constructing a low-rank constrained fully connected layer.

To improve generation and overcome the lack of labeled samples, a deep multiview learning is suggested [50] in the HSI classification task. Nonlocal-GCN [51] utilizes the whole HSI (both labeled and unlabeled data) as input and trains

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

the GCN in a semi-supervised way. S2FL [52] constructs a self-learning architecture with multimodal RS data to learn meaningful features.

As mentioned in Section I, DL-based methods have led to impressive progress but still suffer from two problems: heavy computing complexity and model size. Some recent approaches have turned their attention toward developing efficient models. LDWN [53] adopts 3-D depthwise convolution to reduce parameters in convolution layers and removes the BatchNorm and ReLU layers to restrain overfitting. SC-EADNet [4] proposes an asymmetric dilated convolution block to capture multiscale spatial information efficiently. The 3D-DL method [54] is introduced to fuse spatial and spectral information at a lower computation cost.

However, the performance of these lightweight models is not as good as that of the heavy models. Therefore, we introduce SRP and AMF to boost the performance of deep models.

B. Structural Reparameterization

Multibranch networks usually exhibit superior performance over plain networks. SRP is an attractive technique and recently has been applied in image classification [55], [56], detection [57], and segmentation [58]. The SRP technique is proposed to train a multibranch network and then convert it into a plain network. Before the conversion, a multibranch convolution layer can be written formally as

$$O_{i,j} = \sum_{b}^{B} \sum_{k,l,m}^{K_b^h, K_b^w, M} W_{b,k,l,m} \cdot F_{i+k-1, j+l-1, m} \tag{1}$$

where O is the output and F is the input, i, j represent the pixel location, and k, l are the spatial indices of the convolution weights. B represents the branch number, and $W_b \in \mathbb{R}^{K_b^h, K_b^w, M, N}$ is a convolution kernel in the bth branch, M, N are the input channel and output channel, respectively. Here, the output is the sum of all the branches and the bias item is omitted. In the evaluation stage, these convolution kernels are summed up to a single kernel $\dot{W} \in \mathbb{R}^{K_{\max}^h, K_{\max}^w, M, N}$, $(K_{\max}^h, K_{\max}^w)$ are the maximum kernel size among W

$$\dot{W} = \sum_{b}^{B} W_b. \tag{2}$$

C. Extended Morphological Profiles

Morphological operators are used to extract morphological spatial information and build morphological profiles on a spectral channel. The fundamental operators are dilation and erosion. Dilation Δ and erosion ∇ operators are defined as follows, respectively,

$$(F \Delta E)(i, j) = \max(F(i-k, j-l) | (k, l) \in E) \tag{3}$$

$$(F \nabla E)(i, j) = \min(F(i-k, j-l) | (k, l) \in E) \tag{4}$$

where F is the input feature/image and E is a pixel set that can be defined by the structural element. i, j and k, l are the coordinate of input feature F and structural element E, respectively. EMP takes PCA to reduce the spectral channel first and then adopt two composite morphological operators, opening $(\Delta(\nabla))$ and closing $(\nabla(\Delta))$, as a spatial feature extractor. The opening/closing result is concatenated to get final feature representation.

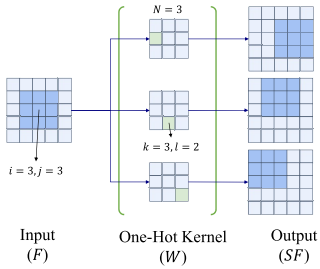


Fig. 1. Visualization of the shift operation. Convolution kernel is defined in one-hot format. Here, the output channel N is three.

III. Proposed Method

This section describes our AMF module and demonstrates how AMF reparameterizes the convolution as a morphology operation. Additionally, we investigate multilayer AMFs and demonstrate that multilayer AMFs are equivalent to the decomposition of structural elements. Finally, we provide the whole architecture of our AMNet and the loss function.

A. Adaptive Morphology Filter

In this section, we introduce the construction and reparameterization procedure of AMF. The binary structural elements in morphological operations can be viewed as a local spatial index set. Therefore, we introduce the shift operation to facilitate end-to-end training of the SE. This operation moves pixels from local index positions to a central position, simplifying subsequent processes. In our work, shift operation can be implemented by a group conventional layer with one-hot type kernels. Assuming that the input of AMF is F, the shift result at position i, j can be defined as

$$\mathrm{SF}(i, j, n) = \sum_{k,l}^{S,S} W_{k,l,n} \cdot F_{i+k-1, j+l-1} \tag{5}$$

where S defines the kernel size, and W represents a one-hot kernel, and N is the output channel for each input, or the number of shift operations. This processing can be implemented by group convolution. The shift process is illustrated in Fig. 1. As a morphology operation, AMF also consists of two basic types: dilation AMF_D and erosion AMF_E. Their structural element E [as referenced in (3) and (4)] can be defined as a group of shift kernels as follows:

$$E = \{W_1, W_2, \ldots, W_N\}. \tag{6}$$

The output channel N is a hyperparameter, and the larger the N, the greater the overparameterization property. In our design, the structural element should be learnable. However, the one-hot type kernel (shift operation) is non-differentiable and cannot be optimized through back-propagation. To overcome this shortcoming, the Gumbel-Softmax estimator is introduced to generate a soft version $\hat{W}$. For training, we initialize $\hat{W}$ with zeros, except for the center value, which is set

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

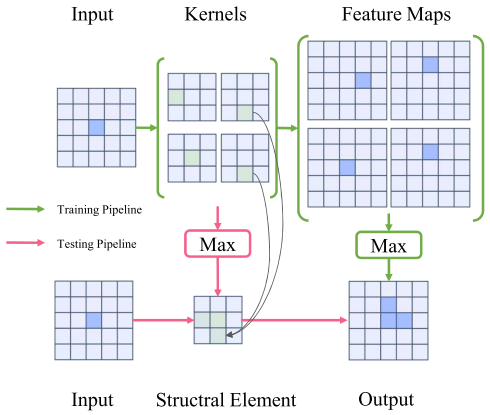


Fig. 2. Reparameterization of AMF. In this case, S is set to three. Note that some convolution kernels in the training pipeline are identical, but have no effect on the final outcome. The duplicate kernels are removed by a max operation (black solid arrow).

to one. Then, we can relax it as

$$G_{k,l} = -\log(-\log(U_{k,l})) \tag{7}$$

$$\hat{W}_{k,l} = \frac{e^{(\hat{W}_{k,l}+G_{k,l})/\tau}}{\sum_{k,l}^{S,S} e^{(\hat{W}_{k,l}+G_{k,l})/\tau}} \tag{8}$$

where G is a noise vector that has the same size as $\hat{W}$, and $U_{i,j} \sim U(0,1)$ is sampled from uniform distribution. In the back-propagation, we calculate the gradient according to (8). Meanwhile, in the feed-forward, we convert $\hat{W}$ to W by

$$W = \begin{cases} 0, & i,j \neq \text{argmax}_{i,j}(\hat{W}) \\ 1, & \text{otherwise.} \end{cases} \tag{9}$$

During the testing phase, since there is no need for Gumbel-softmax to propagate gradients, we can utilize the SRP to reparameterize multiple shift operations into a single morphological filter. According to (3)–(5), AMF_D and AMF_E can be described as

$$\text{AMF}_D(i,j) = \max_N \sum_{k,l}^{S,S} W_{k,l,n} \cdot F_{i+k-1,j+l-1} \tag{10}$$

$$\text{AMF}_E(i,j) = \min_N \sum_{k,l}^{S,S} W_{k,l,n} \cdot F_{i+k-1,j+l-1}. \tag{11}$$

Now, we will construct our AMF for the test phase. The N kernels can be reparameterized as a single structural element $\widehat{\text{SE}}$

$$\widehat{\text{SE}} = \max_N(W_n). \tag{12}$$

The procedure for the reparameterization of AMF is shown in Fig. 2. Previous research [42] has shown that the shape of SE is important for extracting spatial information. In our work, the shape is learnable and defined by N shift operations. For one $S \times S$ SE, there are S^2 binary elements. As one shift operation corresponds to one element, N must be at least S^2. Note that repeated kernels are allowed because the max operation can remove duplicates. In our experiment, due to the decomposition strategy, N is set to 9, the specifics are addressed in Section IV-D.

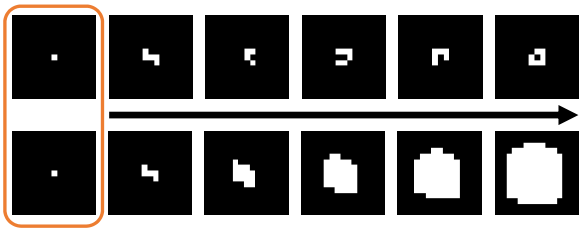


Fig. 3. Structural elements in $L = 5$ stack AMFs (upper) and corresponding large structural elements (lower). The images in the orange box are the initialized weights. The weights of the five layers are trained on the IP dataset. The solid black arrow represents the layers from first to fifth.

B. Adaptive Structural Element Decomposition

Structural elements determine the spatial region where AMF works, and the HSI categorization process benefits greatly from the diverse spatial information. In our work, the SE of AMF is learnable, while the receptive field is fixed. For extracting multiscale spatial information, different sizes of AMF should be taken into account. However, as previously stated, the SE size S determines the model size, which grows exponentially with S. A single 32×32 AMF requires 1024 shift operations at least, and it is unwise to directly optimize so many convolution kernels.

On the other hand, the proposed AMF operates in a depth-wise manner, with consistent input and output channels and no inter-channel information exchange. Based on the chain-rule structural elements decomposition (SED) [59], a large-scale binary AMF can be decomposed into a stack of smaller AMFs. Similar to the convolution kernel decomposition idea proposed by Visual Geometry Group (VGG) [60], SED effectively reduces the parameters/computational overhead of large-scale AMFs. Concurrently, AMFs of varying depths possess receptive fields of different sizes, thereby offering multiscale morphological features, which further enhance the network's performance. The dilation SED and erosion SED are depicted below, respectively,

$$F \Delta E = (((F \Delta E_1) \Delta E_2), \ldots) \Delta E_L \tag{13}$$

$$F \nabla E = (((F \nabla E_1) \nabla E_2), \ldots) \nabla E_L \tag{14}$$

where E represents large $S \times S$ SE and $E_l, l = (1, 2, \ldots, L)$ is 3×3 SEs. Our stack AMFs are adaptive for the dataset. Some kernels of stack AMFs and their corresponding large SEs are shown in Fig. 3. We can discern that the large SE tends to take on a circle shape as the AMF layer goes deeper. The reason is that the network requires isotropic components to be rotation-resistant.

In practice, we implement the shift operation of AMF by group convolution. The group number is equal to the input channel C, and the output channel is set to $N \times C$. After convolution, it will be split into C groups. For each group, we take the maximum/minimum value along the channel dimension. Such operations introduce nonlinearity into the network. Actually, it is the equivalent of the Maxout [61] activation function for AMF_D and the Minout for AMF_E. Those Maxout/Minout results only have one channel. We concatenate them to obtain the input of the next AMF. The smallest size needed to extract spatial information is 3×3. According to the formula for convolutional receptive field,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

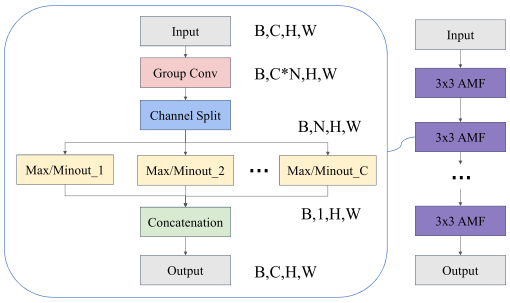


Fig. 4. Visualization of the kernels of stack AMF.

we have $R_l = R_{l-1} + (s-1) \times d$, R_l represents the receptive field of lth layer AMF, the size (s) and stride (d) of small AMFs are 3 and 1, respectively. Therefore, $R_l = 3+2\times(l-1)$. For a single large $S \times S$ AMF, the receptive field is S, i.e., $R_L = S$. Consequently, the layer of stacking AMFs is given by $L = [(S-3)/2]+1$. The details of AMF are exhibited in Fig. 4.

C. Adaptive Morphology Network

For HSI $I \in \mathbb{R}^{H\times W\times D}$, where H and W define the image height and width, and D represents the number of spectral channels. Unlike RGB images, HSI data has hundreds of channels. Many works [5], [62], [63] take PCA to reduce redundant spectral channels, yielding $\tilde{I} \in \mathbb{R}^{H\times W\times P}$, where P is the channel number after dimension reduction. Many studies have highlighted the equivalence between PCA and 1×1 convolution, in our work, we adopt the 1×1 convolution in place of PCA to reduce the spectral channel to P and avoid the additional preprocessing steps [64], [65]. As the HSI classification task needs to attach a label at the pixel level, for a certain pixel $\tilde{I}_{i,j}$ at location i, j, image patch $C_{i,j} \in \mathbb{R}^{K\times K\times P}$ is cropped from $\tilde{I}$, where K represents the patch size.

These cropped patches are initially passed through 1×1 convolution layers to extract shallow features. The output channel is set to $P/2$. Afterward, shallow features will be split into two parts and traverse two parallel modules: the dilation path and the erosion path. Dilation path stacks $3 \times 3\,\mathrm{AMF}_D$ and erosion path stacks $3 \times 3\,\mathrm{AMF}_E$, both of them are described in Section III-B. The layer number in each path is determined by the patch size K. To complete coverage spatial information, the maximum size of morphology operation S is equal to K. Finally, we concatenate the first 1×1 convolution result, shallow features, and all the hierarchy output into two paths to acquire rich, multiscale features. These features will be sent to a full connection (FC) layer to get classification results. The entire pipeline is shown in Fig. 5. Notably, we can find all components of EMP in our AMFNet, like spectral reduction, multiscale morphology operations, and the concatenation feature. Therefore, our AMNet can be considered as a DL version of EMP.

Our AMNet is a full convolutional segmentation network [66], and FC is implemented via 1×1 convolution layers in practice. In contrast to traditional classification architecture, the output of AMNet has the same spatial size as the input. In the training phase, we use cropped patches to train our model and ignore the non-labeled and test samples. Because the input size of AMNet can be non-fixed, we evaluate our model with full-size HSI data after reparameterization. The loss is shown as follows:

$$\mathrm{Loss} = -Y \times \log(\mathrm{softmax}(\mathrm{logist} \times M) \tag{15}$$

where the logist is the output of AMFNet, and Y is the one-hot format label. M is the training label mask. If pixel i, j is background or belongs to the test dataset, $M_{i,j} = 0$, otherwise; $M_{i,j} = 1$.

TABLE I
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR INDIAN PINES DATASET

Class ID	Class Name	Train	Test	Total
1	Alfalfa	5	41	46
2	Corn-notill	5	1423	1428
3	Corn-mintill	5	825	830
4	Corn	5	232	237
5	Grass-pasture	5	478	483
6	Grass-trees	5	725	730
7	Grass-pasture-mowed	5	23	28
8	Hay-windrowed	5	473	478
9	Oats	5	15	20
10	Soybean-nottill	5	967	972
11	Soybean-mintill	5	2450	2455
12	Soybean-clean	5	588	593
13	Wheat	5	200	205
14	Woods	5	1260	1265
15	Buildings-Grass-Trees-Drives	5	381	386
16	Stone-Steel-Towers	5	88	93
Total		80	10169	10249

TABLE II
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR SALINAS DATASET

Class ID	Class Name	Train	Test	Total
1	Brocoli-green-weeds-1	5	2004	2009
2	Brocoli-green-weeds-2	5	3721	3726
3	Fallow	5	1971	1976
4	Fallow-rough-plow	5	1389	1394
5	Fallow-smooth	5	2673	2678
6	Stubble	5	3954	3959
7	Celery	5	3574	3579
8	Grapes-untrained	5	11266	11271
9	Soil-vinyard-develop	5	6198	6203
10	Corn-senesced-green-weeds	5	3273	3278
11	Lettuce-romaine-4wk	5	1063	1068
12	Lettuce-romaine-5wk	5	1922	1927
13	Lettuce-romaine-6wk	5	911	916
14	Lettuce-romaine-7wk	5	1065	1070
15	Vinyard-untrained	5	7263	7268
16	Vinyard-vertical-trellis	5	1802	1807
Total		80	54049	54129

IV. EXPERIMENTS AND RESULTS

A. Datasets

In this section, we conduct experiments with five extensively used datasets: Indian Pines (IP), Salinas (SA), Pavia University (PU), Houston2013 (Houston13) and Houston2018 (Houston18). Tables I–V show detailed partition information for each dataset.

1) *Indian Pines:* This dataset was captured by AVIRIS sensors over the Indian Pines forest site in Indiana, USA.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

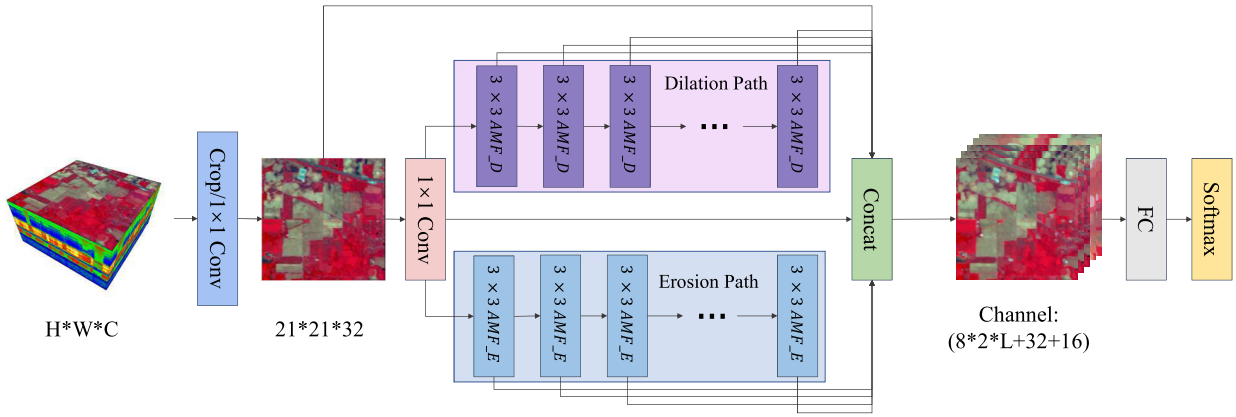


Fig. 5. Pipeline of AMFNet, L is the number of layers for both the dilation and erosion path. In the test phase, all AMF modules will be reparameterized to the corresponding morphology operation.

TABLE III
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR PAVIA UNIVERSITY DATASET

Class ID	Class Name	Train	Test	Total
1	Asphalt	5	6626	6631
2	Meadows	5	18644	18649
3	Gravel	5	2094	2099
4	Trees	5	3059	3064
5	Painted-Metal-Sheets	5	1340	1345
6	Bare-Soil	5	5024	5029
7	Bitumen	5	1325	1330
8	Self-Blocking-Bricks	5	3677	3682
9	Shadows	5	942	947
Total		45	42731	42776

TABLE IV
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR HOUSTON2013 DATASET

Class ID	Class Name	Train	Test	Total
1	Healthy-grass	5	1251	1256
2	Stressed-grass	5	1254	1259
3	Synthetic-grass	5	697	902
4	Trees	5	1244	1249
5	Soil	5	1242	1247
6	Water	5	325	330
7	Residential	5	1268	1273
8	Commercial	5	1244	1249
9	Road	5	1252	1257
10	Highway	5	1227	1232
11	Railway	5	1235	1240
12	Parking-Lot 1	5	1233	1238
13	Parking-Lot 2	5	469	474
14	Tennis-Court	5	428	432
15	Running-Track	5	660	665
Total		75	14954	15029

TABLE V
CLASS NAME AND SAMPLE NUMBER OF EACH CLASS FOR HOUSTON2018 DATASET

Class ID	Class Name	Train	Test	Total
1	Healthy-grass	5	9794	9799
2	Stressed-grass	5	32497	32502
3	Artificial-turf	5	679	684
4	Evergreen-trees	5	13590	13595
5	Deciduous-trees	5	5016	5021
6	Bare-earth	5	4511	4516
7	Water	5	261	266
8	Residential-buildings	5	39767	39772
9	Non-residential-buildings	5	223747	223752
10	Roads	5	45861	45866
11	Sidewalks	5	34024	34029
12	Crosswalks	5	1513	1518
13	Major-thoroughfares	5	46343	46348
14	Highways	5	9860	9865
15	Railways	5	6932	6937
16	Paved-parking-lots	5	11495	11500
17	Unpaved-parking-lots	5	141	146
18	Cars	5	6542	6547
19	Trains	5	5364	5369
20	Stadium-seats	5	6819	6824
Total		100	504756	504856

It has 200 spectral bands and a total of 21 025 pixels are subdivided into 16 categories of various plants, while the remaining pixels represent the background. There is a clear disparity in the number of distinct categories.

2) *Salinas:* This dataset was acquired by a high-resolution imaging spectrometer over the Salinas Valley and has 204 spectral bands with a spatial resolution of 3.7 m per pixel. The size of the data is 512 × 217 pixels, divided into 9 categories. Each category includes a nonuniform number of pixels, from 900 to 10 000.
3) *Pavia University:* This dataset comprises 115 contiguous spectral bands with a size of 610 × 340 captured over Pavia, Italy. In spite of having up to 2 207 400 pixels, only 42 776 samples from nine categories are labeled.
4) *Houston2013:* The University of Houston campus dataset was collected by the National Center for Airborne Laser Mapping (NCALM) for the 2013 GRSS data fusion contest. The HSI data consists of 144 bands, with each band measuring 349 × 1905 pixels. The ground truth data comprises 15 029 samples across 15 different land cover classes.
5) *Houston2018:* This dataset was publicly released for the 2018 data fusion contest, containing an image with 48 spectral bands and spatial dimensions of 601 × 2384. The ground truth map was generated from a high-quality image with dimensions of 1202 × 4768, and was used to resize the original HSI image and sample ground truth

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON INDIAN PINES DATASET. PARA MEANS PARAMETERS. K INDICATES A KILO AND M INDICATES A MILLION

Class	Tradition		Heavy				Lightweight					Ours
	SVM	EMP	2D-CNN	SSUN	ASSMN	HybridSN	M3D-CNN	MSRN	LDWN	MFORMER	LMSS	(train/test)
OA(%)	40.45	63.80	48.27	60.33	52.77	64.20	49.12	64.67	71.28	60.31	67.08	**77.62**
AA(%)	51.18	71.48	63.65	71.65	68.90	78.38	59.10	78.35	80.86	72.93	80.39	**86.99**
Kappa(%)	33.10	58.86	42.77	54.96	47.50	60.27	43.11	60.83	67.62	55.73	63.38	**74.80**
Para(K)	/	/	2,660	1,216	3513	5,122	36	52	51.71	200	58	24/**11**
FLOPs(M)	/	/	29.98	4.56	1213	250	21.00	3.58	56.31	35.67	15.52	8.87/**3.16**
1	85.37	90.24	97.56	97.56	90.24	100.00	73.17	100.00	100.00	100.00	100.00	100.00
2	13.91	31.90	64.09	49.68	27.06	52.71	27.27	48.00	68.73	47.93	53.41	60.33
3	10.30	55.03	37.39	40.73	41.70	45.70	17.58	61.58	44.43	34.67	57.70	79.44
4	31.03	25.43	37.45	60.34	78.88	89.66	60.78	87.50	79.74	93.97	96.98	98.92
5	37.24	19.87	76.72	58.79	64.64	63.60	53.56	74.69	69.67	70.29	70.71	84.62
6	41.10	88.69	32.22	83.45	48.28	73.52	38.48	57.38	56.55	51.45	67.72	92.92
7	95.65	100.00	26.62	91.30	91.30	100.00	86.96	100.00	100.00	100.00	100.00	100.00
8	20.93	99.15	71.04	98.94	99.79	95.98	89.64	96.41	99.37	92.81	100.00	98.60
9	80.00	100.00	86.67	100.00	80.00	100.00	93.33	100.00	100.00	100.00	100.00	100.00
10	48.40	45.60	35.16	35.99	49.22	61.22	45.60	65.96	52.12	55.95	56.05	54.18
11	51.18	70.04	36.69	57.92	41.59	47.88	49.51	54.08	50.90	53.63	57.63	71.22
12	27.04	45.24	61.05	46.60	48.30	66.16	45.24	56.12	66.33	50.17	65.14	74.30
13	98.00	99.50	91.50	94.00	93.50	100.00	85.00	92.50	91.00	88.50	96.00	99.75
14	73.49	98.87	79.37	80.24	71.51	85.08	74.76	77.06	90.00	84.13	80.48	97.41
15	6.30	74.02	73.75	50.92	78.74	83.99	59.32	90.29	76.38	70.60	93.44	80.21
16	98.86	100.00	79.55	100.00	97.73	88.64	45.45	92.05	92.05	72.73	90.91	100.00

TABLE VII

EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON SALINAS DATASET. PARA MEANS PARAMETERS. K INDICATES A KILO AND M INDICATES A MILLION

Class	Tradition		Heavy				Lightweight					Our
	SVM	EMP	2D-CNN	SSUN	ASSMN	HybridSN	M3D-CNN	MSRN	LDWN	MFORMER	LMSS	train/test
OA(%)	71.68	78.27	56.97	79.47	80.65	81.36	78.92	85.51	84.83	82.34	83.63	**89.92**
AA(%)	76.71	82.40	64.26	85.04	87.18	84.74	82.12	88.04	87.95	85.37	88.56	**94.52**
Kappa(%)	68.34	75.51	53.14	77.24	78.38	79.45	76.71	83.97	83.17	80.41	81.94	**88.78**
Para(K)	/	/	2660	1217	3516	4846	36	52	53	203	58	24/**11**
FLOPs(M)	/	/	29.98	4.56	1213	51.33	21.33	3.58	57.45	36.39	15.55	8.93/**3.21**
1	99.70	99.80	75.90	56.44	96.21	63.87	74.60	72.75	87.28	76.30	80.99	100.00
2	51.17	93.95	76.05	83.15	95.00	100.00	96.40	96.24	97.07	96.45	99.30	97.30
3	27.70	7.01	55.86	88.43	92.95	99.59	87.01	99.59	95.23	88.08	83.56	97.71
4	97.19	99.21	74.95	99.21	99.71	95.97	94.10	98.56	97.84	93.23	99.21	95.00
5	98.50	98.20	68.76	91.62	98.17	93.30	94.58	98.05	97.27	96.97	95.59	98.96
6	96.36	96.23	56.88	90.87	92.82	77.59	90.44	85.53	94.33	80.35	81.08	98.59
7	99.50	99.44	54.00	66.84	74.71	78.74	92.03	82.51	93.62	88.25	76.44	96.23
8	87.55	96.42	36.71	65.74	76.98	58.61	54.96	68.78	68.20	64.95	55.68	79.88
9	98.35	98.92	56.70	94.55	95.19	98.29	91.76	99.48	94.11	94.79	98.56	99.25
10	2.99	57.80	69.54	92.06	84.54	74.95	80.45	82.55	87.47	87.17	90.35	93.04
11	95.39	94.07	86.45	98.31	76.58	68.67	52.49	63.41	59.55	63.50	85.42	99.11
12	99.53	100.00	68.00	99.84	94.54	97.14	93.03	93.13	99.64	97.24	99.74	100.00
13	98.90	99.23	71.57	90.67	95.17	80.79	66.08	89.90	94.73	92.97	84.96	93.39
14	87.61	96.43	76.53	89.20	95.31	99.91	87.51	100.00	83.10	83.85	100.00	99.90
15	10.33	1.84	52.44	62.07	33.03	86.48	71.64	86.48	74.35	76.61	91.73	67.95
16	76.53	79.86	48.72	91.62	94.01	81.91	86.79	91.73	83.35	85.29	94.28	95.06

data. The dataset consists of 20 classes and a total of 504 856 samples.

B. Experimental Setup

In this section, we delineate four aspects of the experiments conducted on the five datasets. The codes can be found from the website https://github.com/zhu-xlab/Adaptive-Morphology-Filter.

1) *Optimization:* In the experiments, as suggested by [4] and [50], five samples of each category are selected randomly for the training set, while the remaining samples are used for the testing phase. The parameters of the network are optimized with the Adam optimizer with default parameter settings. The cosine annealing learning rate is adopted to facilitate a smooth training process, and the iteration count is set to 4000. Additionally, we reduced the spectral channel dimension to $P = 32$ through the first convolution layer on all datasets. For the patch size of the comparative methods, we adopt the settings as specified in their original papers.
2) *Evaluation Metrics:* For further evaluation, we employ general evaluation metrics such as overall accuracy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON PAVIA UNIVERSITY DATASET. PARA MEANS PARAMETERS. K INDICATES A KILO AND M INDICATES A MILLION

Class	Tradition		Heavy				Lightweight					Our
	SVM	EMP	2D-CNN	SSUN	ASSMN	HybridSN	M3D-CNN	MSRN	LDWN	MFORMER	LMSS	train/test
OA(%)	67.84	72.32	54.25	64.40	60.70	68.51	61.60	73.33	69.87	66.47	77.63	**78.17**
AA(%)	71.62	76.88	45.85	73.47	65.69	63.54	68.14	72.34	79.58	66.09	78.02	**87.48**
Kappa(%)	58.25	64.19	4.007	55.96	49.78	59.95	53.59	66.42	63.69	57.64	71.31	**72.84**
Para(K)	/	/	2657	1197	3439	4845	20	51	31	149	55	19/**6**
FLOPs(M)	/	/	29.98	4.51	1213	51.33	10.15	3.57	8.50	21.61	14.81	7.50/**1.79**
1	56.63	59.79	47.18	56.97	32.51	64.11	53.65	63.99	82.31	56.28	53.05	78.11
2	77.68	77.22	72.81	61.72	70.25	70.74	55.59	74.27	49.46	72.00	81.54	68.05
3	29.56	50.33	41.79	50.81	58.60	59.31	56.16	46.13	80.23	76.07	90.69	92.75
4	97.71	96.96	39.03	97.12	93.00	16.35	76.66	59.63	85.03	71.59	68.16	95.58
5	98.88	99.85	57.09	99.48	99.25	83.43	90.52	93.66	98.88	88.43	92.24	99.98
6	18.21	40.51	24.86	44.29	38.38	96.50	72.87	89.17	87.76	51.55	91.88	93.85
7	90.87	83.02	62.34	78.57	64.60	90.94	88.53	97.66	95.09	81.58	94.19	94.35
8	75.31	84.58	34.24	72.45	45.04	75.74	62.52	81.86	95.49	61.98	78.41	66.53
9	99.79	99.68	33.33	99.79	89.60	14.76	56.79	44.69	41.93	35.35	52.02	98.04

TABLE IX

EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON HOUSTON13 DATASET. PARA MEANS PARAMETERS. K INDICATES A KILO AND M INDICATES A MILLION

Class	Tradition		Heavy				Lightweight					Our
	SVM	EMP	2D-CNN	SSUN	ASSMN	HybridSN	M3D-CNN	MSRN	LDWN	MFORMER	LMSS	train/test
OA(%)	47.55	68.10	34.31	63.48	65.05	42.06	62.02	57.37	56.53	58.08	60.35	**83.45**
AA(%)	52.93	71.68	38.56	65.53	69.37	48.19	65.42	61.22	60.86	60.31	65.23	**86.41**
Kappa(%)	43.77	65.72	29.22	60.62	62.32	37.59	59.01	54.00	53.12	54.72	57.18	**82.12**
Para(K)	/	/	2660	1207	3472	4846	28	51	40	155	57	22/**9**
FLOPs(M)	/	/	29.98	4.53	1213	51.33	14.76	3.58	40.27	25.64	15.11	8.08/**2.37**
1	76.08	80.18	43.50	88.36	80.82	38.12	80.58	56.02	68.86	53.45	62.76	89.93
2	47.32	85.83	30.02	63.49	96.56	45.48	47.32	52.44	51.48	49.16	25.78	99.51
3	99.13	96.39	69.51	94.94	100.00	97.69	97.98	93.93	93.06	97.40	95.95	99.99
4	90.07	94.27	15.82	93.70	57.38	23.57	72.56	44.47	64.25	62.47	56.01	98.88
5	80.27	96.77	42.04	87.07	89.41	59.82	82.94	89.17	84.07	85.37	87.23	99.94
6	81.88	78.44	60.00	92.81	97.50	84.38	68.75	80.00	77.19	66.56	85.94	98.81
7	11.48	66.51	18.13	57.17	54.63	39.43	58.75	55.50	66.75	72.68	74.35	82.36
8	12.99	28.65	20.10	33.33	25.67	26.47	57.55	33.49	51.33	52.30	30.91	57.88
9	72.49	59.34	13.79	72.81	71.45	23.42	41.30	34.72	31.84	55.89	48.02	75.79
10	0.00	34.62	54.17	49.10	56.22	59.49	57.77	73.24	58.18	59.66	72.26	67.14
11	5.04	59.27	19.43	43.09	41.38	26.99	37.24	45.77	7.15	31.30	55.45	60.09
12	13.27	29.23	43.97	16.53	25.98	15.39	41.12	49.19	32.08	28.34	43.24	70.38
13	17.46	69.61	44.40	26.08	46.55	64.01	73.71	72.41	69.61	61.21	80.60	95.71
14	95.98	96.69	64.54	71.87	97.87	73.75	84.87	79.43	83.92	70.69	99.76	100.00
15	90.53	99.39	38.93	92.52	99.08	44.89	78.93	58.47	73.13	58.17	60.00	99.66

(OA), average accuracy (AA), and kappa coefficient (Kappa). OA calculates the ratio of correctly classified samples in all samples, AA is the average of the accuracy of each class, and Kappa represents the classification accuracy. To limit the influence of random initialization, each experiment is conducted ten times, and the mean is calculated. In addition, to evaluate the complexity, we compared the number of model parameters and floating-point operations per second (FLOPs) with other DL-based methods.

3) *Comparison With Related Works:* To verify the validity and performance of our proposed method, we compare it with some classical and recent methods. These methods include two traditional methods (SVM [34], EMP [41]), four heavy CNN-based methods (2D-CNN, SSUN [67], ASSMN [68], HybridSN [69]), and five lightweight CNN-based methods (M3D-CNN [70], RMSN [71], LDWN [53], LMSS [72], and MFORMER [30]). SVM classifies each pixel according to the spectral information. EMP is a traditional method that makes comprehensive use of both spectral and spatial information. 2D-CNN uses the spatial information of the image band for classification. M3D-CNN proposes a 3-D convolution block to extract spectral and spatial features synchronously. SSUN adopts the LSTM module to retain the context information to obtain the 3-D features. ASSMN method constructs a spectral–spatial multiscale network to achieve multiscale feature perception and contextual information. HybridSN designs a kind of network structure to mix 2-D convolution and 3-D convolution. RMSN blends multiscale depthwise convolution to generate rich features in lightweight way. LDWN takes the 3-D depthwise convolution to learn spectral–spatial features with few parameters.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

TABLE X
EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON HOUSTON18 DATASET. PARA MEANS PARAMETERS. K INDICATES A KILO AND M INDICATES A MILLION

Class	Tradition		Heavy				Lightweight					
	SVM	EMP	2D-CNN	SSUN	ASSMN	HybridSN	M3D-CNN	MSRN	LDWN	MFORMER	LMSS	Ours
OA(%)	24.10	38.50	18.45	30.05	35.06	19.89	42.12	39.57	47.13	38.50	37.22	**53.18**
AA(%)	42.10	59.47	39.12	56.59	56.43	48.84	53.24	57.89	62.66	58.19	63.01	**72.50**
Kappa(%)	19.41	32.60	11.77	25.30	27.51	15.80	33.68	31.61	39.28	31.77	31.40	**46.00**
Para(K)	/	/	2662	1192	3402	4847	19	52	19	78	57	20/**7**
FLOPs(M)	/	/	29.98	4.49	1213	51.33	4.24	3.58	12.80	8.45	14.41	6.73/**1.04**
1	79.75	84.68	17.00	84.39	76.59	32.32	70.58	50.65	61.71	59.99	38.02	88.50
2	66.03	83.01	24.13	63.28	60.77	42.77	49.17	53.88	55.88	64.16	58.44	78.96
3	100.00	95.73	94.40	100.00	99.56	99.85	83.06	100.00	100.00	100.00	100.00	99.88
4	62.48	94.94	26.76	78.34	90.50	31.60	80.70	30.07	78.45	55.95	68.48	97.52
5	46.29	70.00	27.35	41.53	61.40	23.35	41.55	41.73	53.03	48.76	44.22	72.12
6	17.87	66.30	36.29	77.01	87.83	80.87	81.89	90.80	89.60	86.90	86.34	90.94
7	66.28	67.43	60.54	88.51	98.85	91.19	75.48	98.47	98.08	89.27	100.00	99.96
8	50.83	67.07	29.31	51.32	29.62	30.28	53.28	42.37	56.22	57.50	76.29	59.80
9	17.94	29.33	12.64	11.79	27.61	7.47	42.91	39.45	46.50	31.10	21.38	48.80
10	0.00	0.71	10.29	28.87	21.55	10.07	13.70	25.55	30.24	20.17	28.37	20.96
11	0.04	8.93	10.23	21.56	20.65	15.31	15.43	21.21	14.47	10.62	13.72	34.66
12	1.26	0.73	23.46	12.69	28.16	32.25	9.45	19.70	16.52	17.18	26.17	38.07
13	2.51	22.78	8.53	21.41	25.70	10.82	37.59	21.74	36.10	33.49	37.99	40.80
14	81.18	81.64	67.76	56.39	39.08	50.72	64.26	76.02	73.69	80.06	79.97	81.66
15	86.25	92.44	34.84	66.49	68.72	57.65	63.75	70.79	68.71	64.24	76.01	71.30
16	0.00	72.36	17.49	58.09	69.60	40.71	30.85	35.01	46.96	48.40	49.26	88.77
17	100.00	99.29	84.40	95.74	65.25	73.05	68.09	100.00	77.30	78.72	100.00	100.00
18	0.52	39.97	58.45	53.53	32.65	76.89	60.78	67.09	79.30	71.55	76.78	80.84
19	11.99	38.91	54.31	32.35	47.93	80.24	55.13	77.52	76.66	70.84	83.37	65.58
20	45.42	73.81	84.31	88.49	76.62	89.13	67.14	95.69	93.66	74.86	95.29	90.81

TABLE XI
DETAILS OF TRAINING AND INFERENCE TIME COST OVER FIVE DATASETS. S INDICATE A SECOND

train(GPU)	IP	SA	PU	Houston13	Houston18
M3D-CNN	36.35s	44.28s	45.33s	53.10s	55.85s
MSRN	44.89s	46.96s	42.85s	56.51s	56.58s
LDWN	78.48s	66.65s	56.23s	55.02s	70.96s
MFORMER	194.32s	209.31s	173.37s	205.26s	224.44s
LMSS-NAS	282.01s	324.33s	237.71s	378.37s	315.78s
Our	139.13s	146.37s	103.01s	116.83s	95.71s
Inference(CPU)	**IP**	**SA**	**PU**	**Houston13**	**Houston18**
M3D-CNN	7.36s	37.70s	13.28s	7.22s	35.64s
MSRN	1.30s	3.56s	2.95s	1.52s	23.56s
LDWN	87.59s	491s	211.59s	154.66s	1047.32s
MFORMER	46.49s	223.53s	209.38s	59.09s	2039.47s
LMSS-NAS	31.00	114.45s	98.78s	33.179s	1137.68s
Our	0.03s	0.27s	0.69s	2.47s	4.83s

LMSS integrates neural architecture search (NAS) with spectral–spatial attention mechanisms to reduce network complexity. MFORMER combines morphological operators for local feature extraction with the transformer architecture to enhance the interplay of spectral and spatial information. For SSUN, ASSMN, HybridSN, LMSS, and MFORMER, we use the official code or reproduction code of these comparison methods. For RMSN and LDWN, we reproduce the network architecture. The training process of the comparison method adopts the default parameters in their paper. All experiments are implemented on an NVIDIA GTX2080ti and in a Python environment. Through qualitative analysis, we compare the classification metrics with these methods. For these used tensorflow framework (M3D-CNN, SSUN, ASSMN, HybridSN), we take tfprof as the complexity profiler, and for these used pytorch frameworks (2DCNN, RMSN, LDWN, LMSS, MFORMER, Ours), we take thop as the complexity profiler. It is worth noting that different frameworks and performance analyzers might lead to slight variations in model complexity. However, such variations are typically small (e.g., for HyperSN, the parameters of the TensorFlow version amount to 5122.17 K, while for the PyTorch version, the count is 5122.42 K). Therefore, these discrepancies do not impact our conclusions.

4) *Parameter Analysis and Ablation Study:* There is a hyperparameter plays a crucial role in the proposed method: the size of the cropped patches (K) in training. We conduct comparative analysis experiments to investigate the impact of K on the evaluation metrics. We employed two different types of adaptive morphological filters (AMF), namely AMF_E and AMF_D, to construct the Erosion Path and Dilation Path for spatial feature extraction. To better demonstrate the effectiveness of these modules, we conducted corresponding ablation experiments.

C. Classification Results

The evaluation results of different methods on the five datasets are shown in Tables VI–X. These results reveal that, for all five datasets, our proposed AMFNet surpasses other methods on three metrics: OA, AA, and Kappa. For the IP

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

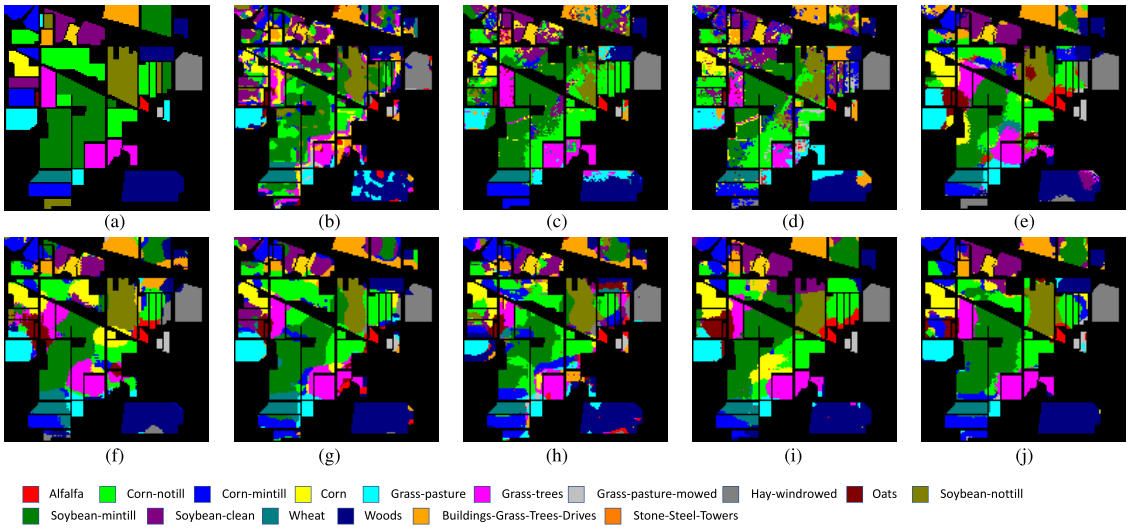


Fig. 6. Ground-truth and classification results for Indian Pines dataset. (a) GT. (b) M3DCNN. (c) SSUN. (d) ASSMN. (e) HybridSN. (f) RMSN. (g) LDWN. (h) MFORMER. (i) LMSS. (j) Our.

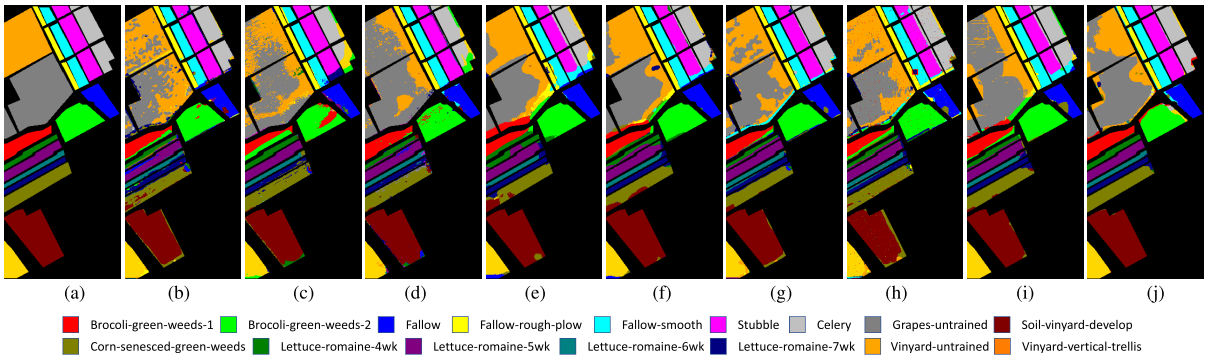


Fig. 7. Ground-truth and classification results for Salinas dataset. (a) GT. (b) M3DCNN. (c) SSUN. (d) ASSMN. (e) HybridSN. (f) RMSN. (g) LDWN. (h) MFORMER. (i) LMSS. (j) Our.

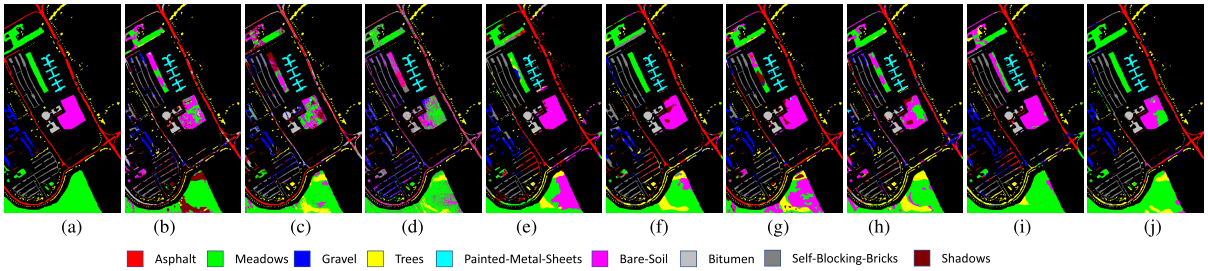


Fig. 8. Ground-truth and classification results for PAVIA University dataset. (a) GT. (b) M3DCNN. (c) SSUN. (d) ASSMN. (e) HybridSN. (f) RMSN. (g) LDWN. (h) MFORMER. (i) LMSS. (j) Our.

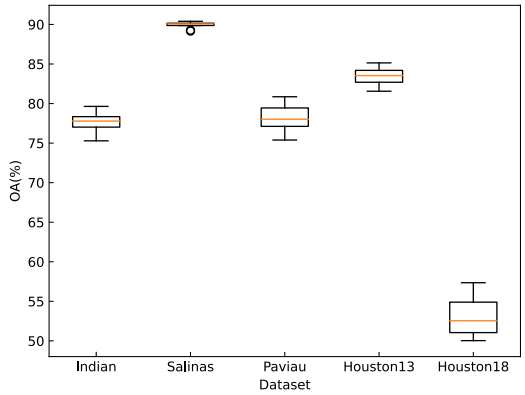


Fig. 9. Statistical analysis of the OA metric of our AMNet over five datasets.

dataset, we reach 77.62% on OA, outperform the second-highest scoring method, LDWN, by approximately 6.34, and outperform the third-highest, LMSS, about 10.54. The ideal outcome is likewise attained for AA and Kappa. For the SA dataset, our AMFNet obtains 89.92% on OA, surpassing the second-highest method, MSRN, by 4.41. Our method's worst results on the SA dataset are 79.88% in the 8th category and 67.95% in the 15th category. The reason is that their spectral curves are similar [73] and hard to distinguish. For the PaviaU dataset, our method achieves the best result on all metrics, for example, we get 78.17% on OA and the second-best method is LMSS reach 77.63%. Similar conclusions can be drawn from the Houston13 dataset, where our approach is the only one to surpass 80% on OA, with EMP being the second-highest method at 68%. On the final and most challenging dataset, Houston18, our approach achieved an OA of 53.18%, surpassing the second-ranked LDWN method by 6.05%. While Houston18's results are lower than other datasets across all

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

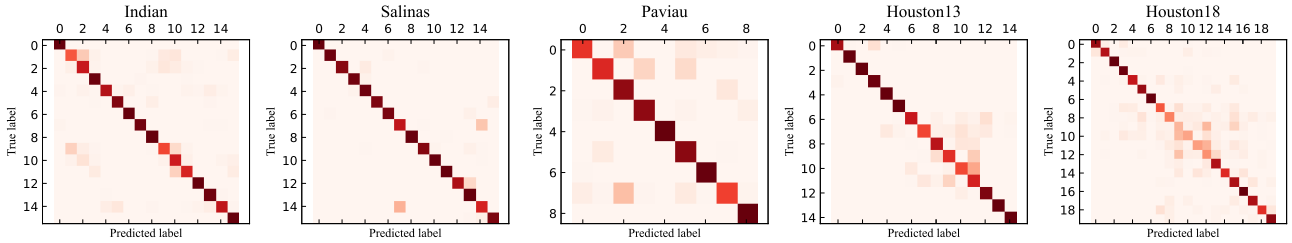


Fig. 10. Confusion matrix of AMFNet on five datasets.

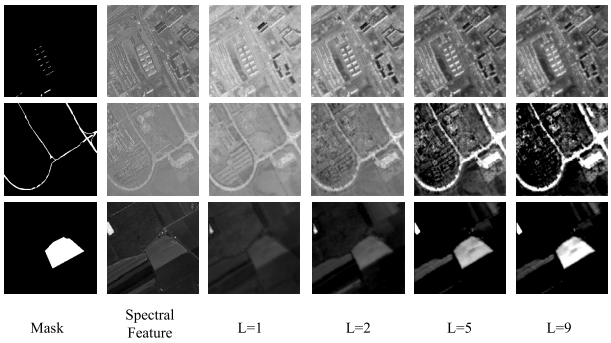


Fig. 11. Visualization of the class activation map, the first and second rows are "Shadows" and "Asphalt" on the PU dataset, respectively, the third row is "Corn-notill" on the SA dataset. Spectral feature means the output of two 1×1 convolutions, and L means the Lth level in erosion path and dilation path.

metrics. We attribute this to three factors: Houston18 has the largest number of classes (20), the least spectral channel (48), and significantly larger volume of data.

Our results for the proposed method are obtained by averaging the results of ten times experiments, and the statistical proofs are shown in Fig. 9. Our method exhibits the largest fluctuation on the Houston18 dataset, with an OA oscillating between 57.35% and 50.03%. However, even the worst result outperforms the second-best method on the Houston18 dataset. Similar conclusions can be drawn from the other four datasets, which demonstrates the effectiveness of our approach.

The confusion matrices are shown in Fig. 10. It can be observed that the most challenging dataset, i.e., Houston18, exhibits the most severe confusion in the building and road classes. For instance, classes 11 (Sidewalks) and 12 (Crosswalks) show significant confusion, possibly due to their spatial proximity and similar spectral distributions. In contrast, for the other datasets, we can see that the confusion matrices are largely in a diagonal matrix form, further indicating the effectiveness of our approach.

We can also find that all the heavy models have poor performance over three metrics. The reason is the lack of training data. As we mentioned in Section IV-B, for each category, only five samples are used for training. For these heavy models, the excessive number of parameters leads to overfitting and hurts performance. By comparison, the traditional method, EMP, achieves a competitive result. For example, EMP exceeds four DL-based methods (2D-CNN, SSUN, ASSMN, and M3D-CNN) with 63.80% on OA for the IP dataset. Especially on the PaviaU and Houston13 datasets, EMP achieves the third and second-best in terms of OA, respectively. We can see that these traditional methods have the advantage in the situation with limited data compared with DL-based methods. And the EMP is an effective feature extraction method in HSI classification tasks. Finally, we can see that AMFNet achieves reliable accuracy on all five datasets with few training samples, which demonstrates the robustness of our method.

It is noteworthy that among the five lightweight approaches, MFORMER outperforms heavy models and traditional methods in most cases. It also surpasses an earlier lightweight method, M3D-CNN, on the IP, SA, and PU datasets. However, it significantly trails other recent lightweight strategies. On one hand, due to the multihead attention mechanism, whose parameter size grows exponentially with the input, MFORMER possesses several times the parameter count compared to other methods, exacerbating the risk of overfitting. On the other hand, the lack of bias inherent in the attention mechanism makes it challenging for MFORMER to learn effective features from a limited dataset.

Another advantage we observe from these tables is that our AMFNet is the most lightweight model in the inference phase among all five datasets. For the Houston18 dataset, spectral channel only involves 48, it reduces the parameters of some models like M3D-Net or LDWN, which heavily rely on the number of spectral bands for their size, but our model is still smaller. We can also find that the SRP significantly reduces the number of parameters. For example, on the Houston18 dataset, AMFNet has about 20 K in parameters during the training phase, which is larger than M3D-CNN (19 K). However, with the structure reparameterization, the model size is decreased by 65% to as little as 7 K without any precise loss.

These tables also list the results of the metric FLOPs. AMFNet requires the lowest calculation burden. For example, it takes only 8.87 M of FLOPs with training for IP dataset. Furthermore, we utilize SRP to further reduce the FLOPs to 3.16 M in the inference stage. Note that both the second-fast MSRN method and our approach have similar computational costs. However, our method, being patch-free, allows for entire HSI image as input instead of splitting it into patches pixel by pixel. This also significantly improves the inference speed. To demonstrate the computational efficiency, we record the running time for training and inference, and the results are presented in Table XI. It can be observed that although the training time of our method is not particularly outstanding compared to other methods, our inference speed significantly surpasses them. It is mainly due to two reasons. First, low computational cost is the benefit of the SPR technique. Second, the patch-free structure of AMFNet enables the model feeding the entire HSI image during inference phase, avoiding redundant input.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

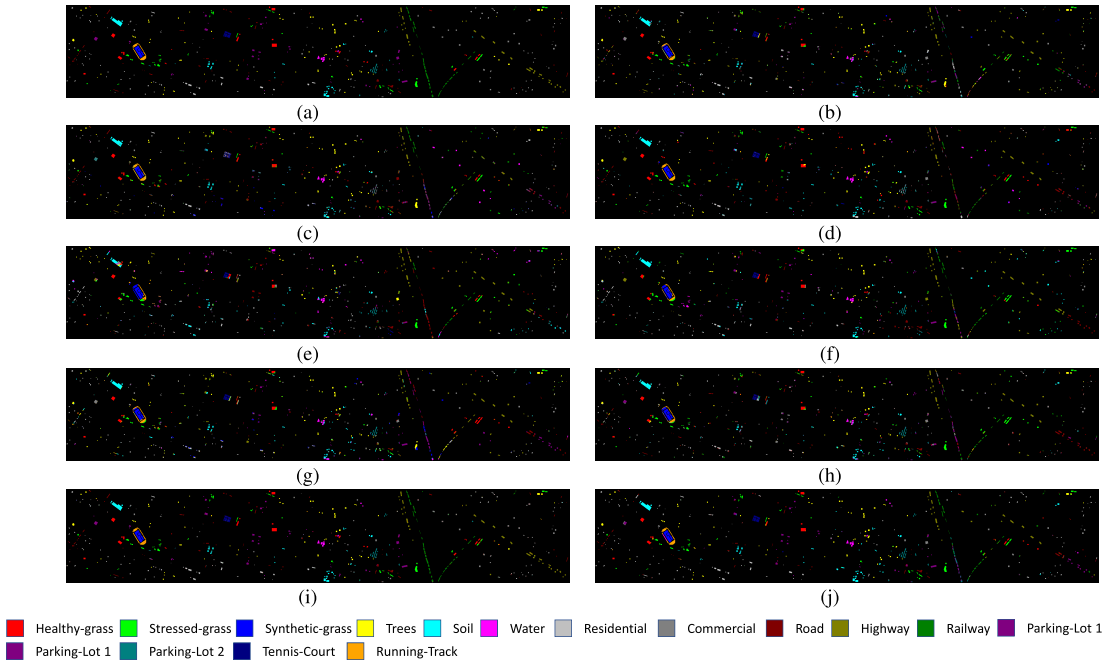


Fig. 12. Ground-truth and classification results for Houston 2013 dataset. (a) GT. (b) M3DCNN. (c) SSUN. (d) ASSMN. (e) HybridSN. (f) RMSN. (g) LDWN. (h) MFORMER. (i) LMSS. (j) Our.

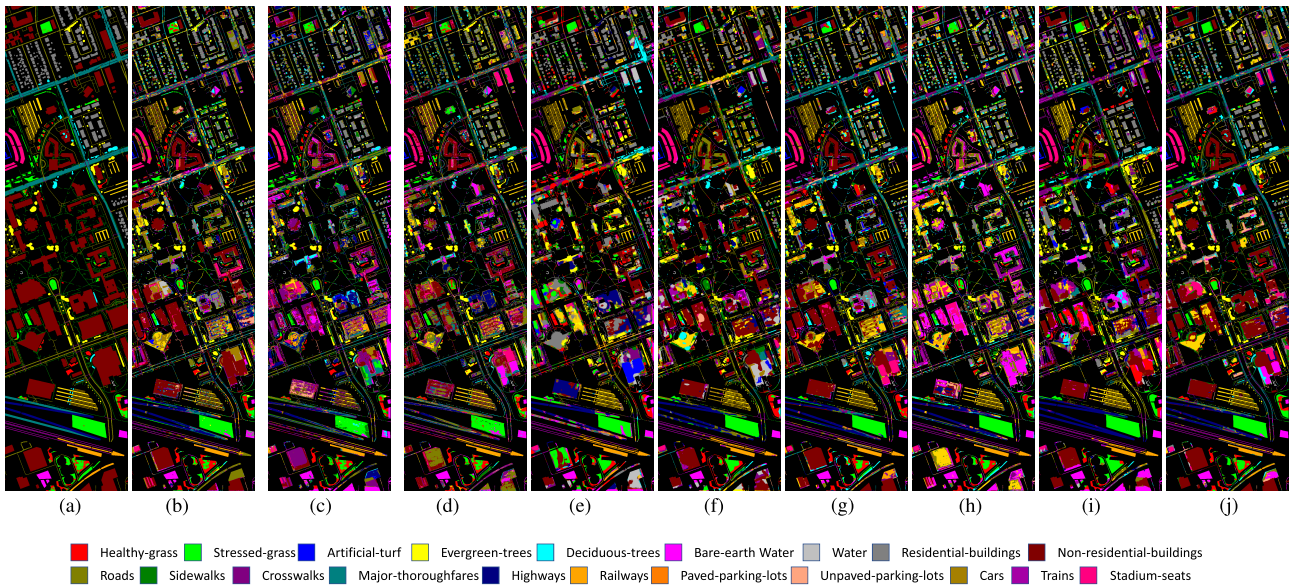


Fig. 13. Ground-truth and classification results for Houston 2018 dataset. (a) GT. (b) M3DCNN. (c) SSUN. (d) ASSMN. (e) HybridSN. (f) RMSN. (g) LDWN. (h) MFORMER. (i) LMSS. (j) Our.

1) Visualization: We conduct qualitative analysis based on the visual classification results. Figs. 6(a), 7(a), 8(a), 12(a), and 13(a) show the ground-truth maps of the five datasets. And the false-color images generated from the classification results of each method are shown from (b) to (j).

Of all the methods, M3DCNN performs the worst, which may be attributed to the minimal spatial size of its input. This leads to unsatisfactory results. A similar issue is observed with MFORMER. The smaller patch size (11×11) hinders its ability to extract broader spatial information, leading to fragmented classification outcomes. The fact that there is no discernible perception gap between our method and state-of-the-art methods, which supports our claim that AMNet can be competitive with state-of-the-art approaches while being significantly lighter.

Furthermore, inspired by methods such as CAM [74], we multiply the weights of specific classifiers with the feature maps to obtain class activation heatmaps, to better understand our AMF. The results are shown in Fig. 11. It can be observed that the pure spectral features contain more noise, while with the stacking of AMF, the feature maps become smoother. Meanwhile, the activation level of the corresponding class gradually increases, resulting in enhanced contrast with the surrounding land cover. This indicates that our AMF effectively integrates spatial information from the surroundings.

Additionally, we employed the t-SNE [75] technique to visualize the features from all lightweight methods on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

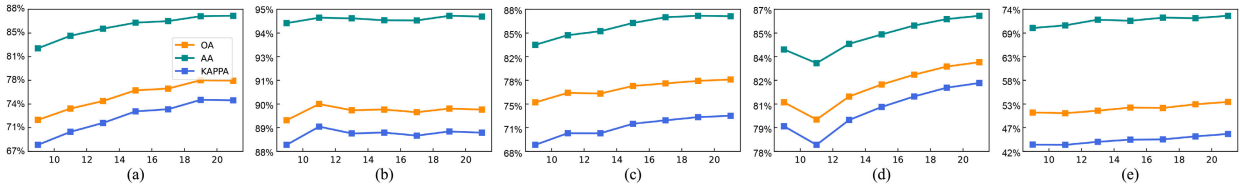


Fig. 14. Sensitivity analysis of the hyperparameters K for five datasets (a) IP, (b) SA, (c) PU, (d) Houston13, and (e) Houston18.

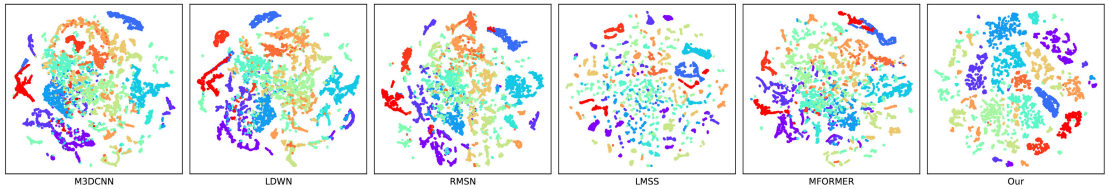


Fig. 15. Visualization results of feature clustering with five comparison lightweight methods. Features extracted from the Houston13 dataset.

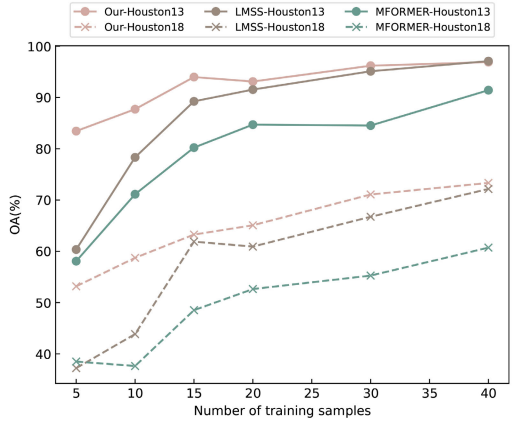


Fig. 16. Classification results of different number of labeled training samples on the Houston13 and Houston18 datasets.

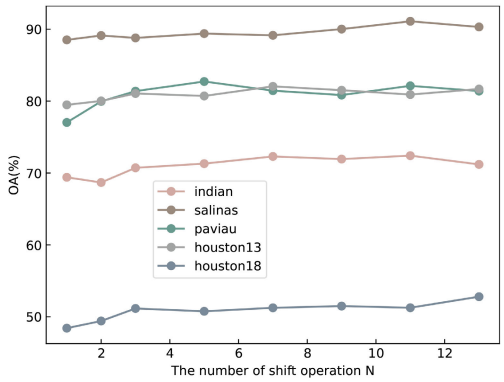


Fig. 17. Classification results of different number of N on the five datasets.

Houston13 dataset, the results of which are depicted in Fig. 15. From the figure, it is evident that, due to the limited training data, the clustering outcomes of almost all comparative algorithms appear scattered. However, our approach demonstrates the smallest inter-class distance and the largest intra-class distance. This observation is consistent with our experimental results, further validating the efficacy of our feature extraction method.

D. Parameter Analysis and Ablation Study

The crop size (K) utilized in the training phase is a crucial hyperparameter in our approach. The hyperparameter determines the spatial information. A small crop size K can capture detailed spatial structure, and a large K can extract features of a smooth area. Note that the layer number of the dilation/erosion path L depends on K, as mentioned in Section III-B. Thus, the receptive field is affected by the hyperparameter K. We designed a series of tests in order to clearly demonstrate the sensitivity of K. We evaluate our model on three metrics (OA, AA, and Kappa) with various K settings on all five datasets. The result is displayed in Fig. 14. We can see that, although there are fluctuations (e.g., for the Houston13 dataset, $10 < K < 12$), larger cropping sizes have higher accuracy. As the computational cost of the network increases exponentially with K, we simply set K as 21.

The number of shift operations, N, serves as another hyperparameter, which dictates the scope of the structural element. To more rigorously ascertain the impact of N on performance, we varied N from 1 to 13 and tested its efficacy across all five datasets. The experimental outcomes are illustrated in Fig. 17. As can be discerned from the figure, our model exhibits insensitivity to the hyperparameter N. With the increment of N, there is a slight enhancement in network performance, which is optimal around values of 5–13. In this work, we set N to 9.

Our network comprises two distinct AMF modules, namely AMF_E and AMF_D. To validate the effectiveness of these two modules, we design three network models: one standard AMFNet, one consisting only of the AMF_E module (erosion path), and one model consisting only of the AMF_D module (dilation path). We evaluate these three models on all five datasets, and each experiment is repeated ten times for average results. The results are shown in Table XII. We can see that the performance declines when either the erosion path or the dilation path is absent. However, the results on the SA dataset show minimal variation, which could be attributed to the dataset's simplicity. The SA dataset exhibits smooth spatial variations and imposes less demand on the spatial feature extractor.

In this article, for each category, we randomly select five training samples. To evaluate the performance of our model with varying training sample sizes, we adjust the number of training samples from 5 to 40 and assess our model on the two most intricate datasets in our experiments, namely Houston13 and Houston18. To further underscore the superiority of our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
ABLATION OA RESULTS OF THE EROSION PATH AND DILATION PATH

Net	IP	SA	PU	Houston13	Houston18
AMFNet	77.62%	89.98%	78.17%	83.45%	53.18%
Only Erosion	75.55%	89.53%	75.78%	82.74%	49.32%
Only Dilation	74.32%	89.98%	75.59%	82.82%	49.48%

approach, we also compare it against two state-of-the-art methods, i.e., LMSS and MFORMER. Classification outcomes are depicted in Fig. 16. As discerned from the figure, our approach performs well with an extremely limited number of training data (5–15), while both LMSS and MFORMER exhibit a noticeable gap with us in performance. As the sample size increases, LMSS gradually approaches our results, but MFORMER consistently lags behind. This discrepancy may be attributed to the transformer structure adopted by MFORMER, whose inherent lack of bias potentially hinders it from extracting effective features in scenarios with scant data.

V. CONCLUSION

In this work, we introduce AMF, an innovative morphological filter for deep HSI classification. AMF is capable of end-to-end training and is equivalent to non-differentiated binary morphological operations that have been shown to be advantageous for HSI processing. The depthwise feature of AMF guarantees the compact design of our model. Furthermore, the group convolution implementation of AMF enables it to be easily reparameterized into a general morphological filter. Such a transformation shrinks the model size sharply in the inference phase without sacrificing accuracy. Additionally, we stack small AMFs to obtain multiscale spatial information and a sufficient receptive field. Further, we demonstrate that stacking depthwise AMFs is equal to structural element decomposition. We propose a lightweight AMNet based on multiscale stack AMFs. Our model has achieved competitive results on five datasets compared with both heavy and lightweight CNNs. Our proposed method still has one limitation, i.e., the size of the receptive field. As our method does not include downsampling modules, on the one hand, the network can only increase the receptive field by stacking more layers. On the other hand, increasing the number of layers inevitably increases the computational complexity, which contradicts the initial motivation of lightweight network designing. In our future work, we will explore downsampling and upsampling modules that preserve morphological features.

ACKNOWLEDGMENT

The authors would like to express their sincere appreciation to the anonymous reviewers for their insightful comments, which have greatly aided them in improving the quality of the article.

REFERENCES

[1] A. Plaza et al., "Advanced processing of hyperspectral images," in *Proc. IEEE Int. Symp. Geosci. Remote Sens.*, Aug. 2006, pp. 1974–1978.

[2] S. Roessner, K. Segl, U. Heiden, and H. Kaufmann, "Automated differentiation of urban surfaces based on airborne hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 7, pp. 1525–1532, Jul. 2001.

[3] M. J. Khan, H. S. Khan, A. Yousaf, K. Khurshid, and A. Abbas, "Modern trends in hyperspectral image analysis: A review," *IEEE Access*, vol. 6, pp. 14118–14129, 2018.

[4] M. Zhu, J. Fan, Q. Yang, and T. Chen, "SC-EADNet: A self-supervised contrastive efficient asymmetric dilated network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519517.

[5] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[6] M. S. Kotzagiannidis and C.-B. Schönlieb, "Semi-supervised superpixel-based multi-feature graph learning for hyperspectral image data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4703612.

[7] S. Hao, W. Wang, Y. Ye, T. Nie, and L. Bruzzone, "Two-stream deep architecture for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2349–2361, Apr. 2018.

[8] D. Zou and Q. Gu, "An improved analysis of training over-parameterized deep neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–10.

[9] A. Jacot, F. Gabriel, and C. Hongler, "Neural tangent kernel: Convergence and generalization in neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, Eds. 2018, pp. 1–10.

[10] Z. Song, S. Yang, and R. Zhang, "Does preprocessing help training over-parameterized neural networks?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds. 2021, pp. 22890–22904.

[11] X. Chang, Y. Li, S. Oymak, and C. Thrampoulidis, "Provable benefits of overparameterization in model compression: From double descent to pruning neural networks," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 8, pp. 6974–6983.

[12] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[13] Z. Ye, C. Li, Q. Liu, L. Bai, and J. E. Fowler, "Computationally lightweight hyperspectral image classification using a multiscale depthwise convolutional network with channel attention," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[14] Z. Meng, L. Jiao, M. Liang, and F. Zhao, "A lightweight spectral–spatial convolution module for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[15] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1577–1586.

[16] M. E. Paoletti, J. M. Haut, N. S. Pereira, J. Plaza, and A. Plaza, "Ghostnet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10378–10393, Dec. 2021.

[17] J. Wang et al., "NAS-guided lightweight multiscale attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8754–8767, Oct. 2021.

[18] L. Wang, T. Zhu, N. Kumar, Z. Li, C. Wu, and P. Zhang, "Attentive-adaptive network for hyperspectral images classification with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505514.

[19] C. Luo et al., "LWCDnet: A lightweight network for efficient cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409816.

[20] J. Wu, C. Leng, Y. Wang, Q. Hu, and J. Cheng, "Quantized convolutional neural networks for mobile devices," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4820–4828.

[21] Y. Gong, L. Liu, M. Yang, and L. Bourdev, "Compressing deep convolutional networks using vector quantization," 2014, *arXiv:1412.6115*.

[22] S. Han, H. Mao, and W. J. Dally, "Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding," 2015, *arXiv:1510.00149*.

[23] H. Wang, C. Qin, Y. Bai, Y. Zhang, and Y. Fu, "Recent advances on neural network pruning at initialization," 2021, *arXiv:2103.06460*.

[24] X. Guo, B. Hou, B. Ren, Z. Ren, and L. Jiao, "Network pruning for remote sensing images classification based on interpretable CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605615.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

[25] Z. Tian, H. Zhao, M. Shu, Z. Yang, R. Li, and J. Jia, "Prior guided feature enrichment network for few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 1050–1065, Feb. 2022.

[26] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surv.*, vol. 53, no. 3, pp. 1–34, Jun. 2020.

[27] Z. Hou, W. Li, L. Li, R. Tao, and Q. Du, "Hyperspectral change detection based on multiple morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507312.

[28] S. K. Roy, B. Chanda, B. B. Chaudhuri, D. K. Ghosh, and S. R. Dubey, "Local morphological pattern: A scale space shape descriptor for texture classification," *Digit. Signal Process.*, vol. 82, pp. 152–165, Nov. 2018.

[29] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[30] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[31] S. K. Roy, R. Mondal, M. E. Paoletti, J. M. Haut, and A. Plaza, "Morphological convolutional neural networks for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8689–8702, 2021.

[32] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[33] X. Lu, W. Zhang, and X. Li, "A hybrid sparsity and distance-based discrimination detector for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1704–1717, Mar. 2018.

[34] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[35] Q. Du and C.-I. Chang, "A linear constrained distance-based discriminant analysis for hyperspectral image classification," *Pattern Recognit.*, vol. 34, no. 2, pp. 361–373, Feb. 2001.

[36] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[37] S. Rajan, J. Ghosh, and M. M. Crawford, "An active learning approach to hyperspectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 4, pp. 1231–1242, Apr. 2008.

[38] L. Zhang, X. Huang, B. Huang, and P. Li, "A pixel shape index coupled with spectral information for classification of high spatial resolution remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 10, pp. 2950–2961, Oct. 2006.

[39] H. Huang, G. Shi, H. He, Y. Duan, and F. Luo, "Dimensionality reduction of hyperspectral imagery based on spatial–spectral manifold learning," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2604–2616, Jun. 2020.

[40] X. Kang, S. Li, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[41] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[42] Z. Y. Lv, P. Zhang, J. A. Benediktsson, and W. Z. Shi, "Morphological profiles based on differently shaped structuring elements for classification of images with very high spatial resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 12, pp. 4644–4652, Dec. 2014.

[43] H. Lee and H. Kwon, "Contextual deep CNN based hyperspectral classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3322–3325.

[44] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[45] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[46] C. Cheng, H. Li, J. Peng, W. Cui, and L. Zhang, "Deep high-order tensor convolutional sparse coding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520611.

[47] C. Cheng, H. Li, J. Peng, W. Cui, and L. Zhang, "Hyperspectral image classification via spectral–spatial random patches network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4753–4764, 2021.

[48] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[49] T. Han, S. Niu, X. Gao, W. Yu, N. Cui, and J. Dong, "Deep low-rank graph convolutional subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533513.

[50] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.

[51] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.

[52] Z. Xue, X. Yu, A. Yu, B. Liu, P. Zhang, and S. Wu, "Self-supervised feature learning for multimodal remote sensing image land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533815.

[53] B. Cui, X.-M. Dong, Q. Zhan, J. Peng, and W. Sun, "LiteDepthwiseNet: A lightweight network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502915.

[54] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[55] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13728–13737.

[56] X. Ding, Y. Guo, G. Ding, and J. Han, "ACNet: Strengthening the kernel skeletons for powerful CNN via asymmetric convolution blocks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1911–1920.

[57] C.-Y. Wang, A. Bochkovskiy, and H.-Y. Mark Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," 2022, *arXiv:2207.02696*.

[58] X. Wang, C. Dong, and Y. Shan, "RepSR: Training efficient VGG-style super-resolution networks with structural re-parameterization and batch normalization," 2022, *arXiv:2205.05671*.

[59] G. Anelli, A. Broggi, and G. Destri, "Decomposition of arbitrarily shaped binary morphological structuring elements using genetic algorithms," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 2, pp. 217–224, Feb. 1998.

[60] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[61] I. J. Goodfellow, D. Warde-Farley, M. Mirza, A. C. Courville, and Y. Bengio, "Maxout networks," in *Proc. ICML*, vol. 28, 2013, pp. 1319–1327.

[62] W. Sun and Q. Du, "Graph-regularized fast and robust principal component analysis for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3185–3195, Jun. 2018.

[63] J. Jiang, J. Ma, C. Chen, Z. Wang, Z. Cai, and L. Wang, "SuperPCA: A superpixelwise PCA approach for unsupervised feature extraction of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4581–4593, Aug. 2018.

[64] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[65] T.-Y. Lin, A. RoyChowdhury, and S. Maji, "Bilinear CNN models for fine-grained visual recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1449–1457.

[66] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5612–5626, Aug. 2020.

[67] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.

[68] D. Wang, B. Du, L. Zhang, and Y. Xu, "Adaptive spectral–spatial multiscale contextual feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2461–2477, Mar. 2021.

[69] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[70] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[71] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.

[72] C. Cao, H. Xiang, W. Song, H. Yi, F. Xiao, and X. Gao, "Lightweight multiscale neural architecture search with spectral–spatial attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505315.

[73] F. Li, D. A. Clausi, L. Xu, and A. Wong, "ST-IRGS: A region-based self-training algorithm applied to hyperspectral image classification and segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 3–16, Jan. 2018.

[74] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[75] L. van der Maaten, "Accelerating t-SNE using tree-based algorithms," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 3221–3245, Jan. 2014.



Fei Zhou received the B.S. and M.S. degrees in electronic science and technology from the Ocean University of China, Qingdao, China, in 2016 and 2019, respectively, where he is currently pursuing the Ph.D. degree with the Department of Computer Science and Technology.

His research interests include machine vision and pattern recognition.



Xin Sun (Member, IEEE) received the Ph.D. degree from the College of Computer Science and Technology, Jilin University, Changchun, China, in 2013.

He is currently an Experienced Humboldt Researcher at Technical University of Munich, Munich, Germany. He did the Post-Doctoral Research with the Department of Computer Science, Ludwig-Maximilians-Universität München, Munich. His research interests include machine learning, data mining, and computer vision.



Chengze Sun received the B.S. and M.S. degrees in electronic science and technology from the Ocean University of China, Qingdao, China, in 2020 and 2023, respectively.

His research interests include machine learning and pattern recognition.



Junyu Dong (Member, IEEE) received the Ph.D. degree from the School of Mathematical and Computer Sciences, Heriot-Watt University, Edinburgh, U.K., in 2003.

He is currently a Professor and the Head of the Department of Computer Science and Technology, Ocean University of China, Qingdao, China. His research interests include machine learning, video analysis, and image processing.



Xiao Xiang Zhu (Fellow, IEEE) received the M.Sc., Dr.-Ing., and Habilitation degrees in signal processing from Technical University of Munich (TUM), Munich, Germany, in 2008, 2011, and 2013, respectively.

She is the Chair Professor for Data Science in Earth Observation at Technical University of Munich (TUM) and was the founding Head of the Department "EO Data Science" at the Remote Sensing Technology Institute, German Aerospace Center (DLR), Weßling, Germany. From 2019 to 2022, she is a co-coordinator of the Munich Data Science Research School (www.muds.de) and the head of the Helmholtz Artificial Intelligence–Research Field "Aeronautics, Space and Transport." Since May 2020, she has been the PI and the Director of the International Future AI Lab "AI4EO–Artificial Intelligence for Earth Observation: Reasoning, Uncertainties, Ethics and Beyond," Munich, Germany. Since October 2020, she has been the Director of the Munich Data Science Institute (MDSI), TUM. She was a Guest Scientist or Visiting Professor at the Italian National Research Council (CNR-IREA), Naples, Italy, Fudan University, Shanghai, China, the University of Tokyo, Tokyo, Japan and University of California, Los Angeles, CA, USA, in 2009 and 2014, 2015, and 2016, respectively. She is currently a Visiting AI Professor at ESA's Phi-Lab. Her main research interests are remote sensing and Earth observation, signal processing, machine learning and data science, with their applications in tackling societal grand challenges, e.g., global urbanization, UN's SDGs and climate change.

Dr. Zhu has been a member of young academy (Junge Akademie/Junges Kolleg) at the Berlin-Brandenburg Academy of Sciences and Humanities and the German National Academy of Sciences Leopoldina and the Bavarian Academy of Sciences and Humanities. She serves on the Scientific Advisory Board in several research organizations, among others the German Research Center for Geosciences (GFZ, from 2020 to 2023) and Potsdam Institute for Climate Impact Research (PIK). She is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, Pattern Recognition and serves as the area editor responsible for special issues of *IEEE Signal Processing Magazine*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:20:28 UTC from IEEE Xplore. Restrictions apply.